

Recap:

January 26, 2025

When do we say vector $\vec{a}$ and $\vec{b}$ are parallel?

Whenever $\vec{a}$ and $\vec{b}$ point in the same or opposite direction.

def: Two non-zero vectors $\vec{a}$ and $\vec{b}$ are parallel if one is a scalar multiple of the other.

e.g. $\vec{a} = (3, -1, 2)$ $\vec{b} = (7.5, -2.5, 5)$ are parallel

$$\begin{aligned}\vec{b} &= k \vec{a} ? \text{ yes} & k(3) &= 7.5 & \leadsto k &= 2.5 \\ & & k(-1) &= -2.5 & \leadsto k &= 2.5 \\ & & k(2) &= 5 & \leadsto k &= 2.5\end{aligned}$$

e.g. $\vec{a} = (3, -1, 2)$ $\vec{b} = (6, -1, 4)$ are not parallel

$$\begin{aligned}\vec{b} &= k \vec{a} ? \text{ no} & k(3) &= 6 & \leadsto k &= 2 \\ & & k(-1) &= -1 & \leadsto k &= 1 \\ & & k(2) &= 4 & \leadsto k &= 2\end{aligned}$$

X all of these need to be equal

Proposition: For any vector $\vec{a}$ and any scalar k , $\|k\vec{a}\| = |k| \|\vec{a}\|$.

def: A unit vector is a vector of length 1.

Proposition: Take a non-zero vector $\vec{a}$, then the unit vector $\hat{a}$ in the direction of $\vec{a}$ is given by $\hat{a} = \frac{1}{\|\vec{a}\|} \vec{a}$

e.g. $\vec{u} = (3, 1, 5, 1)$ $\hat{u} = \frac{1}{6} \vec{u} = (\frac{1}{2}, \frac{1}{6}, \frac{5}{6}, \frac{1}{6})$

$$\|\vec{u}\| = \sqrt{(3)^2 + (1)^2 + (5)^2 + (1)^2} = \sqrt{9 + 1 + 25 + 1} = \sqrt{36} = 6$$

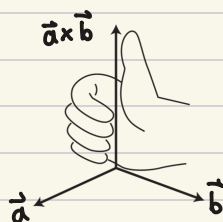
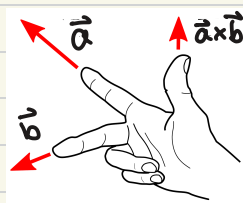
$$\text{proj}_{\vec{u}} \vec{v} = \frac{\vec{u} \cdot \vec{v}}{\vec{u} \cdot \vec{u}} \vec{u} = (\hat{u} \cdot \vec{v}) \hat{u}$$

1.3 The Cross product (which only exists in $\mathbb{R}^3$)

def: Take vectors $\vec{a}, \vec{b}$ in $\mathbb{R}^3$, then $\vec{a} \times \vec{b} \in \mathbb{R}^3$ (the cross product of $\vec{a}$ and $\vec{b}$) is defined by its length and direction:

- $\|\vec{a} \times \vec{b}\| = \|\vec{a}\| \|\vec{b}\| \sin \theta$
- $\vec{a} \times \vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{b}$, and the ordered triple of vectors $(\vec{a}, \vec{b}, \vec{a} \times \vec{b})$ follows the right hand rule.

right hand rule

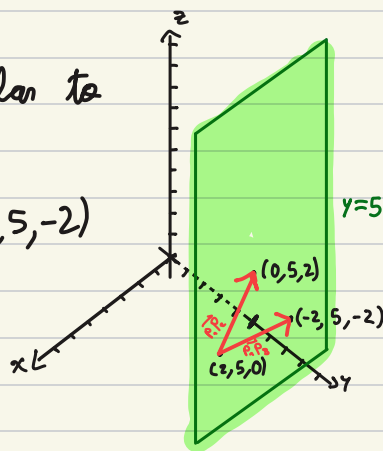


motivation:

- 1) Find a vector $\vec{n}$ that is perpendicular to the plane containing the points

$$P_1(2, 5, 0), P_2(0, 5, 2) \text{ and } P_3(-2, 5, -2)$$

$$\vec{n} = (\overrightarrow{P_1 P_2}) \times (\overrightarrow{P_1 P_3})$$

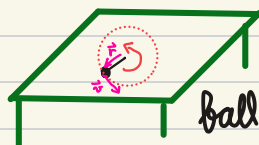
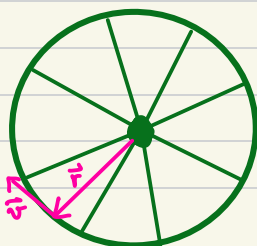


- 2) Can you use a single vector to describe uniform circular motion?

$$\vec{\omega} = \vec{r} \times \vec{v}$$

angular velocity

wheel spinning with constant angular speed



Ball spinning at constant radius with constant speed

Properties: Let $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^3$ and $k \in \mathbb{R}$, then

$$1) \vec{a} \times \vec{b} = -\vec{b} \times \vec{a} \quad (\text{anticommutation})$$

$$2) \vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c} \quad (\text{distributivity})$$

$$3) (\vec{a} + \vec{b}) \times \vec{c} = \vec{a} \times \vec{c} + \vec{b} \times \vec{c}$$

$$4) k(\vec{a} \times \vec{b}) = (k\vec{a}) \times \vec{b} = \vec{a} \times (k\vec{b})$$

e.g.:

$$\begin{aligned} \hat{i} \times \hat{j} &= \hat{k} \\ \hat{j} \times \hat{i} &= -\hat{k} \\ \hat{i} \times \hat{i} &= 0 \end{aligned}$$

$$\begin{aligned} \hat{j} \times \hat{k} &= \hat{i} \\ \hat{k} \times \hat{j} &= -\hat{i} \\ \hat{j} \times \hat{j} &= 0 \end{aligned}$$

$$\begin{aligned} \hat{k} \times \hat{i} &= \hat{j} \\ \hat{i} \times \hat{k} &= -\hat{j} \\ \hat{k} \times \hat{k} &= 0 \end{aligned}$$

Formula for cross product

$$\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}, \quad \vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$$

$$\begin{aligned} \vec{a} \times \vec{b} &= (a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}) \times (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}) \\ &= a_1 \hat{i} \times (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}) + a_2 \hat{j} \times (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}) \\ &\quad + a_3 \hat{k} \times (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}) \\ &= a_1 \hat{i} \times b_2 \hat{j} + a_1 \hat{i} \times b_3 \hat{k} + a_2 \hat{j} \times b_1 \hat{i} + a_2 \hat{j} \times b_3 \hat{k} + a_3 \hat{k} \times b_1 \hat{i} \\ &\quad + a_3 \hat{k} \times b_2 \hat{j} \end{aligned}$$

$$\vec{a} \times \vec{b} = (a_2 b_3 - a_3 b_2) \hat{i} + (a_3 b_1 - a_1 b_3) \hat{j} + (a_1 b_2 - a_2 b_1) \hat{k}$$

e.g. 1) Find a vector $\vec{n}$ that is perpendicular to the plane containing the points

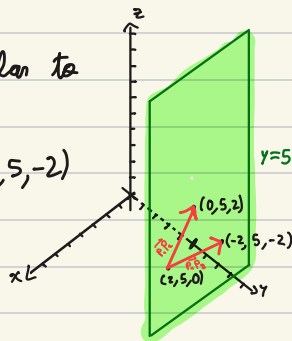
$$P_1(2, 5, 0), P_2(0, 5, 2) \text{ and } P_3(-2, 5, -2)$$

$$\vec{n} = (\vec{P_1 P_2}) \times (\vec{P_1 P_3})$$

$$\vec{P_1 P_2} = (0, 5, 2) - (2, 5, 0) = (-2, 0, 2)$$

$$\vec{P_1 P_3} = (-2, 5, -2) - (2, 5, 0) = (-4, 0, -2)$$

$$\vec{n} = \vec{P_1 P_2} \times \vec{P_1 P_3} = (-2, 0, 2) \times (-4, 0, -2) = (0 - 0) \hat{i} + ((-2)(-4) - (-2)(-2)) \hat{j} + (0 - 0) \hat{k}$$



$$\begin{aligned} \vec{n} &= (-8 - 4) \hat{j} \\ &= -12 \hat{j} \end{aligned}$$